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Rotary actuator

Abstract

A rotary actuator includes a motor, a speed reducer configured to output after reducing a rotation of a rotary shaft of the motor, and a housing accommodating the motor and the speed reducer. The rotary actuator includes a bearing having an inner ring fixed to the rotary shaft and rotatably supporting the rotary shaft, and a cylindrical outer ring fixing portion integrally provided on the housing so as to fix an outer ring of the bearing. A front housing has a jig insertion hole into which an inner ring receiving jig can be inserted, and the jig insertion hole has an opening that overlaps with at least a part of one end surface of the inner ring in view from an axial direction.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4987795	12/1990	Nguyen	N/A	N/A
5911793	12/1998	Kaye	N/A	N/A
2007/0205680	12/2006	Miyoshi et al.	N/A	N/A
2013/0193812	12/2012	Oishi	N/A	N/A
2018/0223973	12/2017	Uematsu	N/A	F16H 25/2015
2019/0207462	12/2018	Uematsu	N/A	H02K 11/33

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
204136005	12/2014	CN	N/A
63-88051	12/1987	JP	N/A
2004-120847	12/2003	JP	N/A
2009-296811	12/2008	JP	N/A
2009296811	12/2008	JP	F16C 27/04
2017-125513	12/2016	JP	N/A
6485365	12/2018	JP	N/A
2019-68518	12/2018	JP	N/A
WO-2018079418	12/2017	WO	F16H 1/32

OTHER PUBLICATIONS

Kimura (WO 2018079418 A1) English Translation. (Year: 2018). cited by examiner
Shibata (JP 2009296811 A) English Translation (Year: 2009). cited by examiner

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of International Patent Application No. PCT/JP2021/010361 filed on Mar. 15, 2021, which designated the U.S. and based on and claims the benefits of priority of Japanese Patent Application No. 2020-049085 filed on Mar. 19, 2020. The entire disclosure of all of the above applications is incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to a rotary actuator.

BACKGROUND

(2) Conventionally, a rotary actuator including a motor and a speed reducer for reducing a rotation of the motor is known.

SUMMARY

(3) An object of the present disclosure is to provide a rotary actuator that is easy to manufacture while increasing and stabilizing a magnetic force of a position detection unit.

(4) A rotary actuator of the present disclosure includes a motor, a speed reducer that outputs after reducing a rotation of a rotary shaft of the motor, a housing containing the motor and the speed reducer, a bearing that has an inner ring fixed to the rotating shaft and rotatably supports the rotary shaft, and a cylindrical outer ring fixing portion that is integrally provided in the housing and fixes the outer ring of the bearing.

(5) The housing has a jig insertion hole into which an inner ring receiving jig can be inserted, and the jig insertion hole has an opening that overlaps with at least a part of one end surface of the inner ring in view from an axial direction.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The above and other objects, features and advantages of the present disclosure will become more apparent from the following detailed description taken in conjunction with the accompanying drawings.

(2) In the drawings:

(3) FIG. 1 is a schematic diagram illustrating a shift-by-wire system to which a rotary actuator of a first embodiment is applied;

(4) FIG. 2 is a cross-sectional view of the rotary actuator of the first embodiment;

(5) FIG. 3 is an enlarged view of Part III of FIG. 2;

(6) FIG. 4 is a plan view of a front housing and a motor side bearing of FIG. 3 as viewed from a IV-IV line;

(7) FIG. 5 is a diagram illustrating a procedure for assembling the motor side bearing of FIG. 3;

(8) FIG. 6 is a diagram illustrating a procedure for assembling a rotating shaft of FIG. 3;

(9) FIG. 7 is a diagram illustrating another procedure for assembling the motor side bearing of FIG. 3;

(10) FIG. 8 is a diagram illustrating a jig insertion hole of the rotary actuator of a second embodiment, and is a diagram corresponding to FIG. 3 of the first embodiment;

(11) FIG. 9 is a plan view of the front housing and the motor side bearing of FIG. 8 as viewed from a IX-IX line;

(12) FIG. 10 is a diagram illustrating a procedure for assembling the rotating shaft of FIG. 8;

(13) FIG. 11 is a diagram illustrating a jig insertion hole of the rotary actuator of a third embodiment, and is a diagram corresponding to FIG. 3 of the first embodiment;

(14) FIG. 12 is a view of the front housing and the motor side bearing of FIG. 11 as viewed from a XII-XII line;

- (15) FIG. 13 is a diagram illustrating a procedure for assembling a rotating shaft of FIG. 12;
- (16) FIG. 14 is a diagram illustrating a jig insertion hole and a seal member of the rotary actuator of a fourth embodiment, and is a diagram corresponding to FIG. 3 of the first embodiment;
- (17) FIG. 15 is a diagram illustrating a jig insertion hole and a seal member of the rotary actuator of a fifth embodiment, and is a diagram corresponding to FIG. 3 of the first embodiment; and
- (18) FIG. 16 is a diagram illustrating a jig insertion hole and a seal member of the rotary actuator of a sixth embodiment, and is a diagram corresponding to FIG. 3 of the first embodiment.

DETAILED DESCRIPTION

(19) In an assumable example, a rotary actuator including a motor and a speed reducer for reducing a rotation of the motor is known. In the rotary actuator, a metal plate is insert-molded into a resin housing. An outer ring of a bearing is press-fitted into the metal plate to be fixed, and a rotary shaft of the motor is press-fitted into an inner ring of the bearing to be fixed. As a result, an amount of variation in a gap between a position detection sensor for a rotor and a magnet is reduced, and a magnetic force acting on the sensor is increased and stabilized.

(20) In the embodiment of fixing both the outer ring and the inner ring of the bearing, it is necessary to receive a force applied to the inner ring when the rotary shaft is press-fitted into the inner ring after the outer ring is press-fitted to be fixed. In a first embodiment, a plurality of points in a circumferential direction of a flat base portion of the metal plate are bent to form an outer ring fixing portion, and the rest of the base portion forms an inner ring receiving portion. Further, in a second embodiment, after the base portion is projected in an axial direction as a whole, the inside of the protruding portion is recessed in an opposite direction. A side surface of the recessed portion forms the outer ring fixing portion, and a bottom surface of the recessed portion forms the inner ring receiving portion. However, both embodiments are difficult to manufacture. In the first embodiment, a holding force of the outer ring is low, and in the second embodiment, there is a concern about cracking of the bearing during manufacturing.

(21) An object of the present disclosure is to provide a rotary actuator that is easy to manufacture while increasing and stabilizing the magnetic force of the position detection unit.

(22) A rotary actuator of the present disclosure includes a motor, a speed reducer that outputs after reducing a rotation of a rotary shaft of the motor, a housing containing the motor and the speed reducer, a bearing that has an inner ring fixed to the rotating shaft and rotatably supports the rotary shaft, and a cylindrical outer ring fixing portion that is integrally provided in the housing and fixes the outer ring of the bearing.

(23) The housing has a jig insertion hole into which an inner ring receiving jig can be inserted, and the jig insertion hole has an opening that overlaps with at least a part of one end surface of the inner ring in view from an axial direction.

(24) By providing such a jig insertion hole, the inner ring can be received by the inner ring receiving jig inserted into the jig insertion hole at the time of assembly. Therefore, it is not necessary to provide the inner ring receiving portion integrally with the outer ring fixing portion. Therefore, by fixing both the outer ring and the inner ring of the bearing, the magnetic force of the position detection unit is increased and stabilized, and the outer ring fixing portion can be made into a simple cylindrical shape, which makes it easy to manufacture.

(25) Hereinafter, a plurality of embodiments of a rotary actuator (hereinafter, referred to as an “actuator”) will be described with reference to the drawings. In the embodiments, components which are substantially similar to each other are denoted by the same reference numerals and redundant description thereof is omitted.

First Embodiment

(26) As shown in FIG. 1, an actuator 10 of the first embodiment is fixed to an outer wall of a case 12 of a vehicle transmission 11 and is used as a power source of a shift-by-wire system 13. The actuator 10 is driven and controlled by a control device 15 in response to a command signal from a shift operation device 14 to operate a shift range switching mechanism 16 of the vehicle

transmission **11** to switch a shift range.

(27) (Actuator)

(28) First, an overall configuration of the actuator **10** will be described with reference to FIG. 2. The actuator **10** includes a housing **20**, a motor **30**, and a speed reducer **40**.

(29) The housing **20** has a cup-shaped front housing **21** and a cup-shaped rear housing **22**. An opening of the front housing **21** and an opening of the rear housing **22** are combined, and the front housing **21** and the rear housing **22** are fastened to each other by bolts **23**. A bottomed cylindrical metal plate **24** is embedded in the front housing **21**. The rear housing **22** has a tubular protrusion **28** that projects to an opposite side of the front housing **21**. A bracket **29** is fixed to an outer wall of the rear housing **22**. The actuator **10** is fixed to the case **12** (see FIG. 1) of the transmission **11** by using the bracket **29**.

(30) The motor **30** has a stator **31** and a rotor **34**. The stator **31** has a stator core **32** fixed to the metal plate **24** by, for example, press fitting, and a coil **33** provided on the stator core **32**. The rotor **34** has a rotary shaft **37** rotatably supported around a rotary shaft center AX1 by a motor side bearing **35** and a speed reducer side bearing **36**, and a rotor core **38** fitted and fixed to an outside of the rotary shaft **37**. The motor side bearing **35** is provided on the metal plate **24**. The speed reducer side bearing **36** is provided on an output member **44**, which will be described later.

(31) The speed reducer **40** includes an eccentric shaft **41**, a ring gear **42**, an eccentric gear **43**, an output member **44**, and a transmission mechanism **45**. The eccentric shaft **41** is provided on an eccentric shaft center AX2 that is eccentric with respect to the rotary shaft center AX1, and is integrally formed with the rotary shaft **37**. The ring gear **42** is provided coaxially with the rotation axis AX1 and is fixed to the rear housing **22**. The eccentric gear **43** has an external tooth portion **47** that meshes with an internal tooth portion **46** of the ring gear **42**, and is supported by a bearing **48** provided on the eccentric shaft **41** so as to allow a planetary motion. The planetary motion is a motion that revolves around the rotation axis AX1 while rotating around the eccentric axis AX2. The rotation velocity of the eccentric gear **43** during planetary motion is changed with respect to the rotation speed of the rotary shaft **37**.

(32) The output member **44** is provided coaxially with the rotation axis AX1 and is rotatably supported by a bearing **49** provided in the rear housing **22**. The transmission mechanism **45** is composed of an engaging protrusion **51** formed in the eccentric gear **43** and an engaging hole **52** formed in the output member **44** into which the engaging protrusion **51** is inserted, and transmits the rotation of the eccentric gear **43** around the eccentric axis AX2 to the output member **44**.

(33) In the actuator **10**, a rotating magnetic field is generated by switching an energizing phase of the coil **33**, and the rotor **34** rotates under the magnetic attraction force or the reaction force generated by the rotating magnetic field. When the eccentric shaft **41** rotates around the rotation axis AX1 together with the rotor **34**, the eccentric gear **43** makes a planetary motion, and the rotation of the eccentric gear **43** decelerated with respect to the rotation of the rotor **34** is output from the output member **44** to the outside.

(34) (Motor Side Bearing)

(35) Next, a configuration around a motor side bearing **35** will be described. In the following description, a direction parallel to a rotation axis AX1 is simply referred to as “axial direction”, and a direction around the rotation axis AX1 is simply referred to as “circumferential direction”.

(36) As shown in FIG. 2, an actuator **10** has a position detection unit **55** that detects a rotational position of a rotor **34**. The position detection unit **55** has a magnetic sensor **56** provided in a front housing **21** and a magnet **57** provided in a rotor core **38**. The magnet **57** is an annular multipole magnet magnetized in the circumferential direction. The magnetic sensor **56** is arranged with a gap with respect to the magnet **57** in the axial direction.

(37) A metal plate **24** has a cylindrical portion **61** into which the stator core **32** is press-fitted, a bottom portion **62** provided at one end of the cylindrical portion **61**, and a cylindrical outer ring fixing portion **63** formed so as to project in the axial direction at a center of the bottom portion **62**.

The metal plate **24** is insert-molded into the front housing **21** made of resin. The outer ring fixing portion **63** is integrally formed with the front housing **21**.

(38) As shown in FIG. **3**, an outer ring **71** of the motor side bearing **35** is press-fitted and fixed to an inside of the outer ring fixing portion **63**. An inner ring **72** of the motor side bearing **35** is press-fitted and fixed to an outside of one end **39** of the rotary shaft **37**. Since both the outer ring **71** and the inner ring **72** of the motor side bearing **35** are fixed in this way, an axial movement of the rotor **34** with respect to the front housing **21** can be suppressed, and an amount of variation in a gap between the magnetic sensor **56** and the magnet **57** can be reduced. Therefore, the gap can be set as small as possible to increase a magnetic force acting on the magnetic sensor **56**, and the magnetic force can be stabilized.

(39) In the embodiment in which both the outer ring **71** and the inner ring **72** of the motor side bearing **35** are fixed as described above, it is necessary to receive a force applied to the inner ring **72** when the rotary shaft **37** is press-fitted into the inner ring **72** after the outer ring **71** is press-fitted and fixed. In the actuator, the outer ring fixing portion and the inner ring receiving portion are integrally provided on the metal plate, but there is a problem that it is difficult to manufacture the metal plate. To solve this problem, in the first embodiment, an inner ring receiving portion is removed from the actuator **10**, and it is configured to receive the force applied to the inner ring **72** by the jig at the time of assembly. A detailed description will be given below.

(40) The outer ring fixing portion **63** is formed in a simple cylindrical shape so as to hold the entire circumference of the outer ring **71**. The inside is open with respect to the outer ring fixing portion **63**, and the metal plate **24** is not provided with a portion for receiving the force applied to the inner ring **72**.

(41) As shown in FIGS. **3** and **4**, the front housing **21** has a jig insertion hole **74** being an opening that overlaps with at least a part of one end surface **73** of the inner ring **72** in view from the axial direction. The jig insertion hole **74** is a through hole provided coaxially with the rotary shaft **37**. An inner diameter **D1** of the jig insertion hole **74** is larger than an inner diameter **D2** of the inner ring **72**. In the present embodiment, the jig insertion hole **74** is formed larger than the one end surface **73**, and has an opening that overlaps the entire one end surface **73** in view from the axial direction.

(42) In a process of assembling the actuator **10**, (1) the outer ring **71** is press-fitted into the outer ring fixing portion **63** as shown in FIG. **5**, and then (2) one end **39** of the rotary shaft **37** is press-fitted into the inner ring **72** while receiving the force applied to the inner ring **72** by an inner ring receiving jig **91** inserted into the jig insertion hole **74** as shown in FIG. **6**. The jig insertion hole **74** is a hole into which the inner ring receiving jig **91** can be inserted.

(43) In FIG. **5**, while pushing the outer ring **71** with a press-fit punch **93**, the outer ring **71** is press-fitted with a constant stroke or until the outer ring **71** reaches a predetermined depth. On the other hand, as shown in FIG. **7**, while pushing the outer ring **71** and the inner ring **72** with a press-fit punch **95**, the outer ring **71** may be press-fitted until the inner ring **72** comes into contact with the inner ring receiving jig **91**.

(44) In FIGS. **5** and **7**, the outer ring **71** is press-fitted while receiving, for example, the bottom side of the front housing **21**. On the other hand, in another embodiment, the outer ring **71** may be press-fitted while receiving the outer ring fixing portion **63** by a jig inserted into the jig insertion hole **74**. As shown in FIG. **3**, the outer ring fixing portion **63** is formed so as to protrude from the bottom portion **62** toward the motor **30**, and a base end portion **75** of the outer ring fixing portion **63** is suitable as a receiving part. The jig insertion hole **74** has the inner diameter **D1** that is larger than the inner diameter **D3** of the outer ring fixing portion **63** and has an opening that overlaps with the base end portion **75** in view from the axial direction.

(45) In FIGS. **5** and **7**, an outer diameter of the inner ring receiving jig **91** is smaller than an inner diameter of the jig insertion hole **74** and the outer ring fixing portion **63**. On the other hand, in another embodiment, a fitting part that can be fitted with the jig insertion hole **74** or the outer ring fixing portion **63** is provided at a position one step lower than the inner ring receiving portion **92** at

the tip of the inner ring receiving jig **91** that receives the inner ring **72** so that the inner ring receiving jig **91** and the inner ring **72** may be positioned using this fitting part. In this configuration, the jig insertion hole **74** or the outer ring fixing portion **63** also functions as a positioning portion.

(46) Returning to FIG. 3, a seal member **81** is attached to the front housing **21** so as to close the jig insertion hole **74**. The seal member **81** has a waterproof and breathable function. That is, the seal member **81** allows gas and vapor to enter and exit the housing **20** through the jig insertion hole **74**, while preventing fixing and liquid from entering and exiting the housing **20** through the jig insertion hole **74**.

(47) The outer ring fixing portion **63** and the front housing **21** have a ventilation hole **76** for communicating a space **A1** on the jig insertion hole **74** side and a space **A2** on the opposite side thereof with respect to the motor side bearing **35**.

(48) A rotation transmission engaging portion **77** is formed at one end **39** of the rotary shaft **37** on the jig insertion hole **74** side. The rotation transmission engaging portion **77** is composed of, for example, a hexagonal hole or the like. The opening of the jig insertion hole **74** overlaps entirely with the rotation transmission engaging portions **77** in view from the axial direction.

(49) (Effects)

(50) As described above, in the first embodiment, the front housing **21** has the jig insertion hole **74** into which the inner ring receiving jig **91** can be inserted, and the jig insertion hole **74** is the opening that overlaps with at least a part of one end surface **73** of the inner ring **72** in view from the axial direction. By providing such a jig insertion hole **74**, the inner ring **72** can be received by the inner ring receiving jig **91** inserted into the jig insertion hole **74** at the time of assembly. Therefore, it is not necessary to provide the inner ring receiving portion integrally with the outer ring fixing portion **63**. Therefore, by fixing both the outer ring **71** and the inner ring **72** of the motor side bearing **35**, the magnetic force of the position detection unit **55** is increased and stabilized, and the outer ring fixing portion **63** can be made into a simple cylindrical shape, which makes it easy to manufacture.

(51) Further, in the first embodiment, the jig insertion hole **74** is formed larger than the one end surface **73**, and has an opening that overlaps the entire one end surface **73** in view from the axial direction. Therefore, the entire surface of one end surface **73** can be received by the inner ring receiving jig **91**.

(52) Further, in the first embodiment, the seal member **81** is provided to prevent the fixing and the liquid from entering and exiting the housing **20** through the jig insertion hole **74**. This makes it possible to prevent foreign matter from entering the housing **20** and increase the internal pressure while providing the jig insertion hole **74**. Further, by providing the jig insertion hole **74** and the seal member **81** in place of the breathing hole and the filter conventionally provided on the transmission **11** side (outer wall of the rear housing **22**) of the housing **20**, since a clearance between the transmission **11** and the housing **20** is expanded, the mountability of the actuator **10** on the transmission **11** can be improved.

(53) Further, in the first embodiment, the outer ring fixing portion **63** and the front housing **21** have a ventilation hole **76** for communicating a space **A1** on the jig insertion hole **74** side and a space **A2** on the opposite side thereof with respect to the motor side bearing **35**. As a result, when an internal pressure of the housing **20** fluctuates, it is possible to breathe through the ventilation hole **76** (air flows between the space **A1** and the space **A2**). Therefore, it is possible to prevent the grease of the motor side bearing **35** from leaking due to breathing through the inside of the motor side bearing **35**, and to improve the life of the motor side bearing **35**.

(54) Further, in the first embodiment, the rotation transmission engaging portion **77** is formed at one end **39** of the rotary shaft **37** on the jig insertion hole **74** side. The opening of the jig insertion hole **74** overlaps entirely with the rotation transmission engaging portions **77** in view from the axial direction. This makes it possible to mechanically rotate the motor **30** from the outside using a

rotating jig that is inserted into the jig insertion hole **74** and engaged with the rotation transmission engaging portion **77**. For example, when the motor **30** fails, the parking lock can be forcibly released.

Second Embodiment

(55) In the following description, a radial direction of the rotating shaft **37** is simply referred to as “radial direction”. In the second embodiment, as shown in FIGS. **8** and **9**, a plurality of jig insertion holes **742** are provided so as to be arranged in the circumferential direction. A width **W1** in the radial direction of the jig insertion hole **742** is larger than a width **W2** in the radial direction of the one end surface **73**. The radial position of the jig insertion hole **742** is substantially the same as the radial position of the one end surface **73**. A part of the opening of the jig insertion hole **742** overlaps a part of one end surface **73** in view from the axial direction.

(56) As shown in FIG. **10**, an inner ring receiving jig **912** has two protruding inner ring receiving portions **922** inserted into the jig insertion hole **742**. An outer diameter of the inner ring receiving portion **922** is substantially the same as an inner diameter of the jig insertion hole **742**. By fitting the inner ring receiving portion **922** into the jig insertion hole **742**, the inner ring receiving jig **912** and the inner ring **72** are positioned. In this configuration, the jig insertion hole **742** also functions as a positioning portion.

(57) According to the actuator **10** of the second embodiment, it is easy to manufacture while increasing and stabilizing the magnetic force of the position detection unit **55** as in the first embodiment.

Third Embodiment

(58) In a third embodiment, as shown in FIGS. **11** and **12**, a plurality of jig insertion holes **743** are provided so as to be arranged in the circumferential direction. In the present embodiment, three jig insertion holes **743** are provided at equal intervals in the circumferential direction. The radial width **W3** of the jig insertion hole **743** is the same as the radial width **W2** of the one end surface **73**. The radial position of the jig insertion hole **743** is substantially the same as the radial position of the one end surface **73**. The entire opening of the jig insertion hole **743** overlaps a part of one end surface **73** in view from the axial direction.

(59) As shown in FIG. **13**, the inner ring receiving jig **913** has two protruding inner ring receiving portions **923** that are inserted into the jig insertion hole **743**. The outer diameter of the inner ring receiving portion **923** is substantially the same as the inner diameter of the jig insertion hole **743**. By fitting the inner ring receiving portion **923** into the jig insertion hole **743**, the inner ring receiving jig **913** and the inner ring **72** are positioned. In this configuration, the jig insertion hole **743** also functions as a positioning portion.

(60) According to the actuator **10** of the third embodiment, it is easy to manufacture while increasing and stabilizing the magnetic force of the position detection unit **55** as in the first embodiment.

Fourth Embodiment

(61) In a fourth embodiment, as shown in FIG. **14**, an annular locking portion **78** protruding inward in the radial direction is provided on the side of the inner wall of the jig insertion hole **744** opposite to the motor side bearing **35**. A seal member **814** has a film **82** having a waterproof and breathable function, a holding portion **83** for holding the film **82**, a claw portion **84** which protrudes from the holding portion **83** into the jig insertion hole **744** and locks to the locking portion **78**, and an O-ring **85** arranged between the holding portion **83** and the front housing **21**. As described above, the seal member **814** may be a snap-fit type.

(62) Further, in the fourth embodiment, the rotary shaft **37** has a ventilation hole **764** for communicating the space **A1** and the space **A2**. The ventilation hole **764** is formed of a groove formed in one end **39** or the like of the rotary shaft **37**. As a result, it is possible to breathe through the ventilation holes **764** when the internal pressure of the housing **20** fluctuates.

Fifth Embodiment

(63) In a fifth embodiment, as shown in FIG. 15, a cylindrical protrusion 65 protruding in the axial direction on the outside of the jig insertion hole 745 is provided on the outer wall of the front housing 21. The seal member 815 has a film 825 having a waterproof and breathable function, a holding portion 835 that holds the film 825 and is press-fitted into the cylindrical protrusion 65, and a cover portion 86 provided so as to cover the film 825 on the outside of the holding portion 835. As described above, the seal member 815 may be an outer wall fixed type.

Sixth Embodiment

(64) In a sixth embodiment, as shown in FIG. 16, the seal member 816 has a film 826 having a waterproof and breathable function and a holding portion 836 that holds the film 826 and is press-fitted into the inner wall of the jig insertion hole 746 As described above, the seal member 816 may be an inner wall fixed type.

(65) The present disclosure is not limited to the embodiments described above, and various modifications are possible within the scope of the present disclosure without departing from the spirit of the invention.

(66) The present disclosure has been made in accordance with the embodiments. However, the present disclosure is not limited to such embodiments and configurations. The present disclosure also encompasses various modifications and variations within the scope of equivalents.

Furthermore, various combination and formation, and other combination and formation including one, more than one or less than one element may be made in the present disclosure.

Claims

1. A rotary actuator, comprising: a motor; a speed reducer configured to output after reducing a rotational speed of a rotary shaft of the motor; a resin housing accommodating the motor and the speed reducer; a bearing having an inner ring press-fitted and fixed to the rotary shaft and configured to support rotatably the rotary shaft; a metal cylindrical outer ring fixing portion integrally provided on the housing by insert molding and to which an outer ring of the bearing is press-fitted and fixed; a magnet provided on a rotor core integrated with the rotary shaft; and a magnetic sensor provided in the housing so as to be arranged with a gap with respect to the magnet in an axial direction parallel to a rotation axis of the rotary shaft and being configured to detect a rotational position of the rotary shaft by a magnetism of the magnet, wherein the housing has a jig insertion hole into which an inner ring receiving jig is inserted, which has an opening overlapping with at least a part of one end surface of the inner ring in view of an axial direction parallel to a rotation axis of the rotary shaft; and the outer ring fixing portion or the rotary shaft has a ventilation hole that communicates a space on a side of the jig insertion hole with respect to the bearing and a space on the side opposite to the jig insertion hole with respect to the bearing.
2. The rotary actuator according to claim 1, further comprising a seal member that prevents solids and liquids from entering and exiting the housing through the jig insertion hole.
3. The rotary actuator according to claim 1, wherein a rotation transmission engaging portion is formed at one end of the rotary shaft on the jig insertion hole side, and the jig insertion hole has an opening that overlaps with the rotation transmission engaging portion in view from the axial direction.
4. A rotary actuator, comprising: a motor including a stator core, a coil provided in the stator core, a rotary shaft, and a rotor core integral with the rotary shaft; a speed reducer configured to output after reducing a rotational speed of the rotary shaft of the motor; a resin housing accommodating the motor and the speed reducer; a bearing having an outer ring and an inner ring press-fitted and fixed to the rotary shaft and configured to support rotatably the rotary shaft; a metal plate having a cylindrical portion into which the stator core is press-fitted, a bottom portion provided at one end of the cylindrical portion, and a cylindrical outer ring fixing portion formed at a center of the cylindrical portion so as to protrude in an axial direction and having both axial ends open for press-

fitting the outer ring, wherein the cylindrical portion and the bottom portion are integrally provided in the resin housing by insert molding; a magnet provided on the rotor core; and a magnetic sensor provided in the housing so as to be arranged with a gap with respect to the magnet in an axial direction parallel to a rotation axis of the rotary shaft and being configured to detect a rotational position of the rotary shaft by a magnetism of the magnet, wherein the housing has a jig insertion hole into which an inner ring receiving jig is inserted, which has an opening overlapping with at least a part of one end surface of the inner ring in view of an axial direction parallel to a rotation axis of the rotary shaft; and the jig insertion hole is formed on an inner side in a radial direction of a bottom holding portion of the housing that holds the bottom portion of the metal plate.
